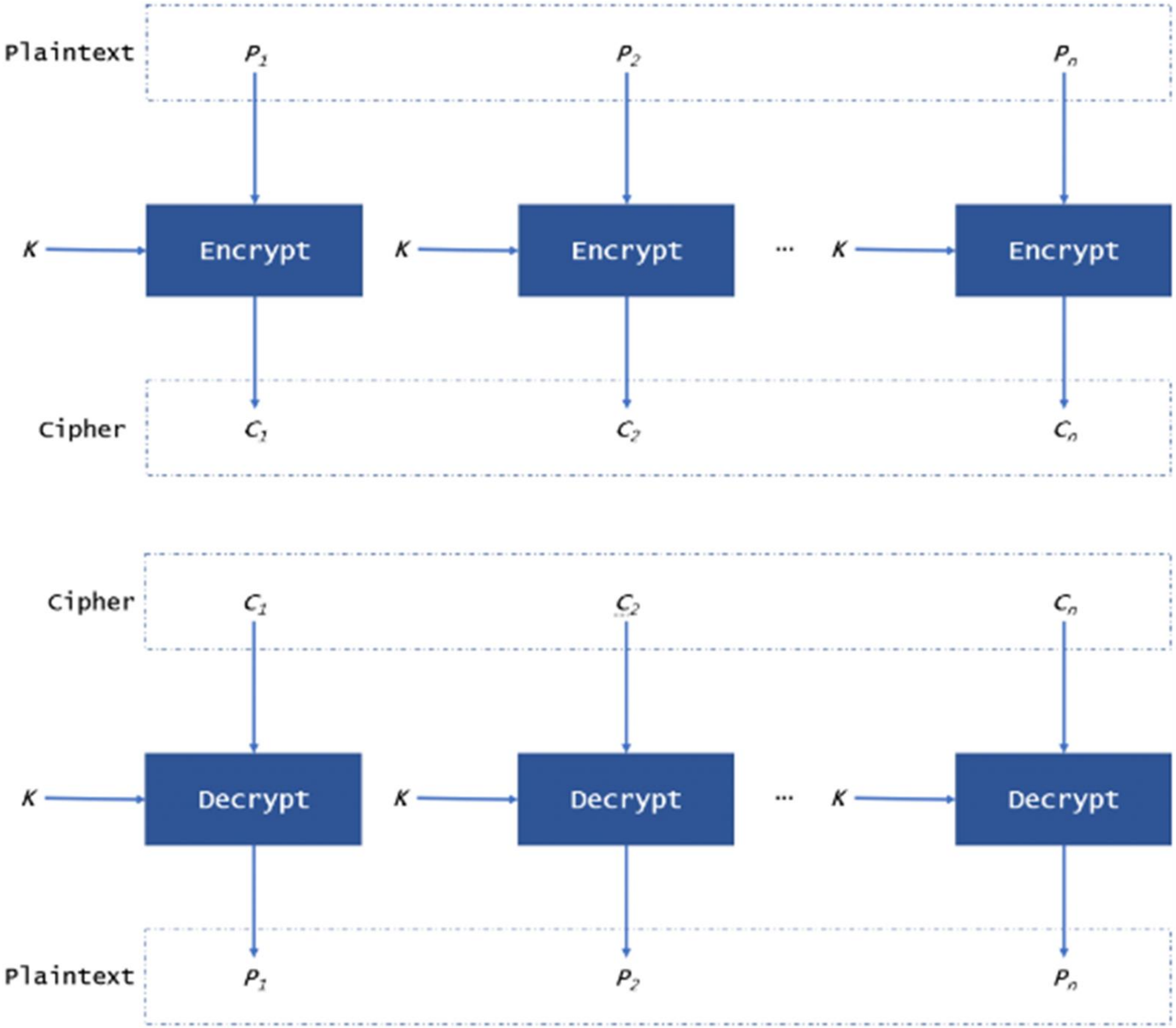
Modes of Operation of Block Ciphers

Symmetric-key encipherment can be done using modern block ciphers. Modes of operation have been devised to encipher text of any size employing either DES or AES.

Topics discussed in this section:

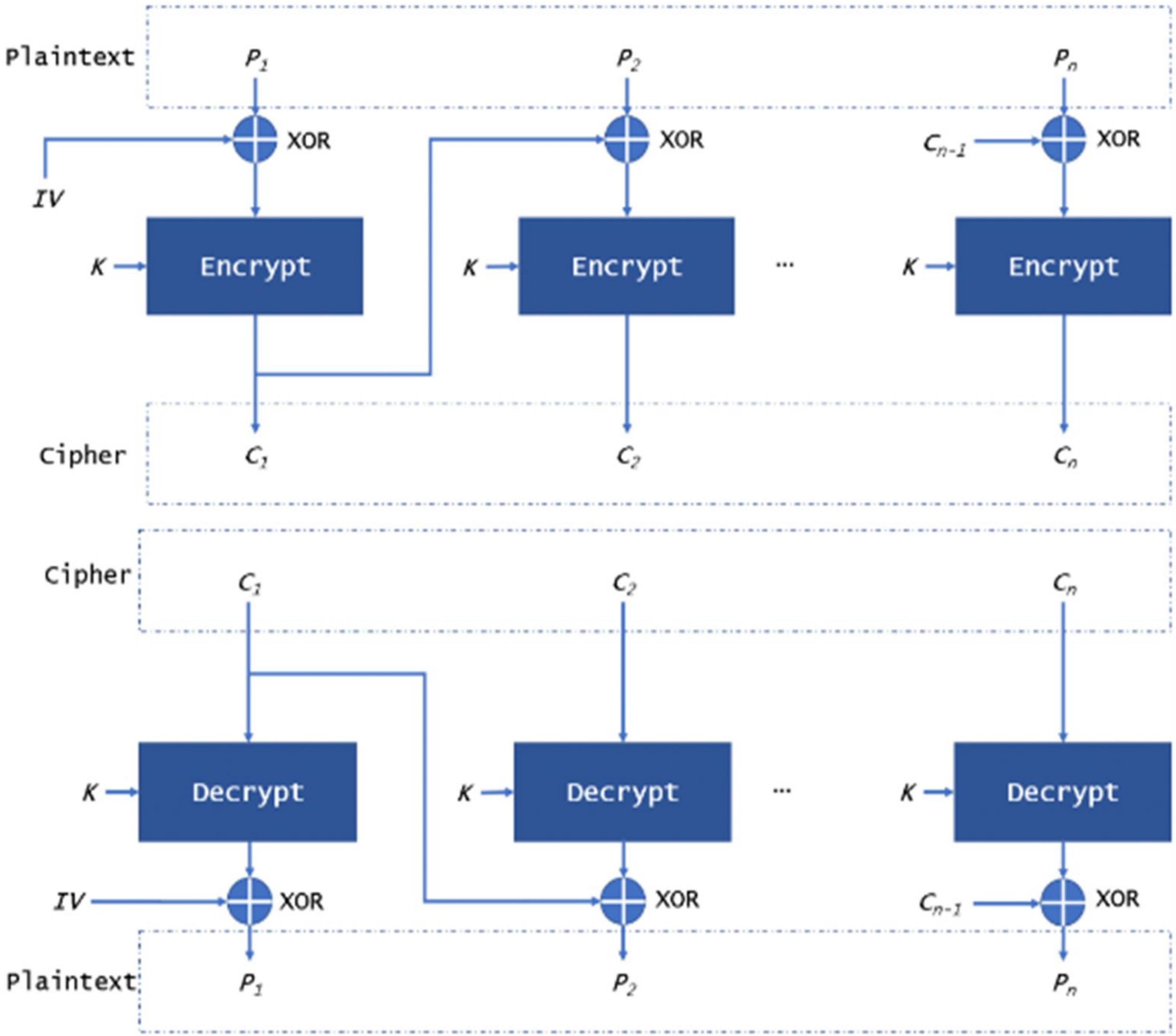
8.1.1 Electronic Codebook (ECB) Mode
8.1.2 Cipher Block Chaining (CBC) Mode
8.1.3 Cipher Feedback (CFB) Mode
8.1.4 Output Feedback (OFB) Mode
8.1.5 Counter (CTR) Mode

The ECB (Electronic Code Book) mode



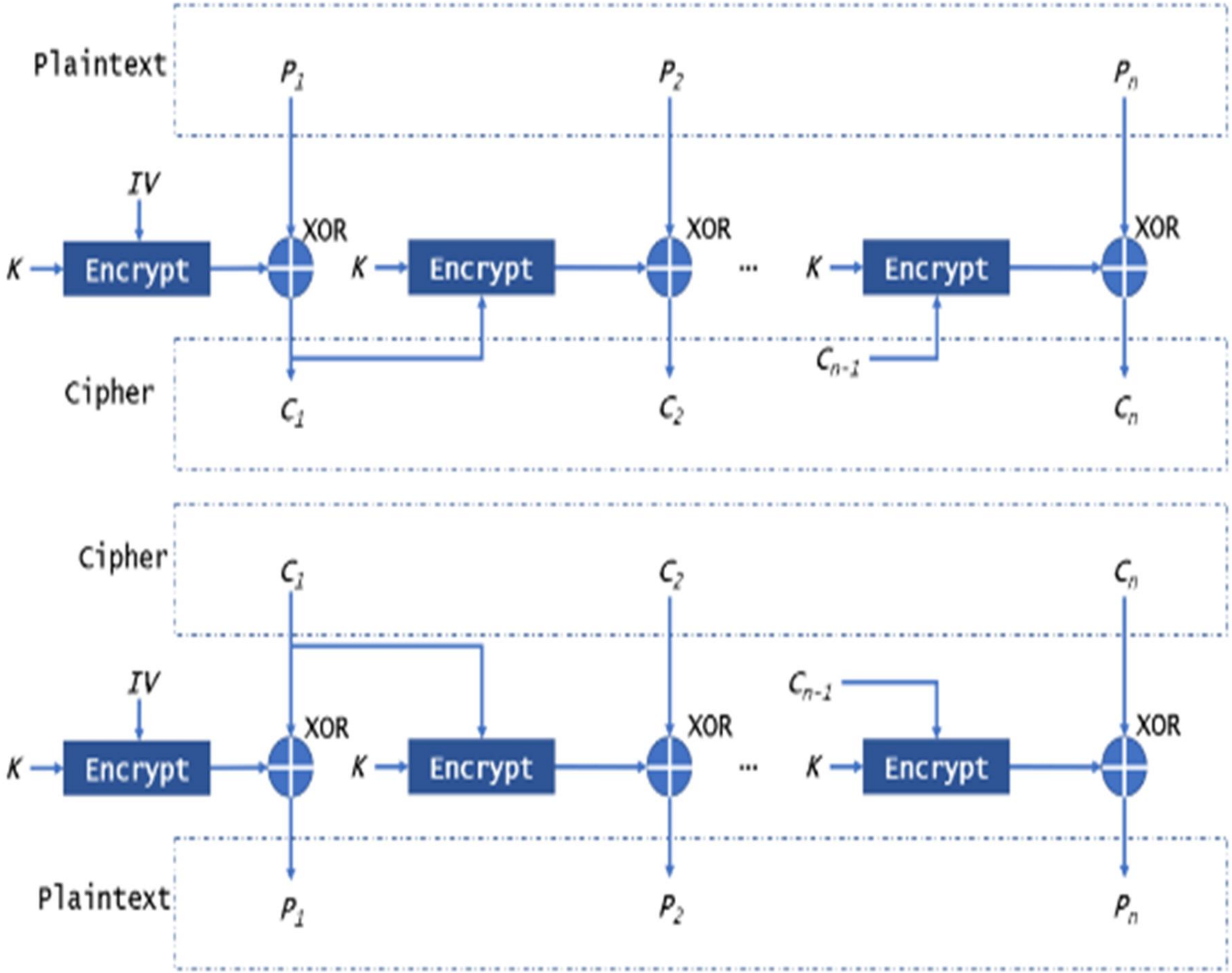
- **Electronic Codebook (ECB):**
- **Description:** Each block of plaintext is independently encrypted.
- **Pros:** Simple and parallelizable.
- **Cons:** Identical plaintext blocks result in identical ciphertext blocks, which can lead to patterns and vulnerabilities.
- **Applications:**
- Image compression: Suitable for lossless compression algorithms.
- Random access encryption of independent blocks.

The CBC (Cipher Block Chaining) mode



- **Cipher Block Chaining (CBC):**
- **Description:** Each plaintext block is XORed with the previous ciphertext block before encryption.
- **Pros:** Provides diffusion, making patterns less noticeable. Suitable for secure communication.
- **Cons:** Requires sequential processing, limiting parallelization. Error propagation in case of a block corruption.
- **Applications:**
- Secure communication protocols (e.g., TLS/SSL): Provides confidentiality and integrity.
- Disk encryption: Helps prevent patterns in encrypted data.

The CFB (Cipher FeedBack) mode

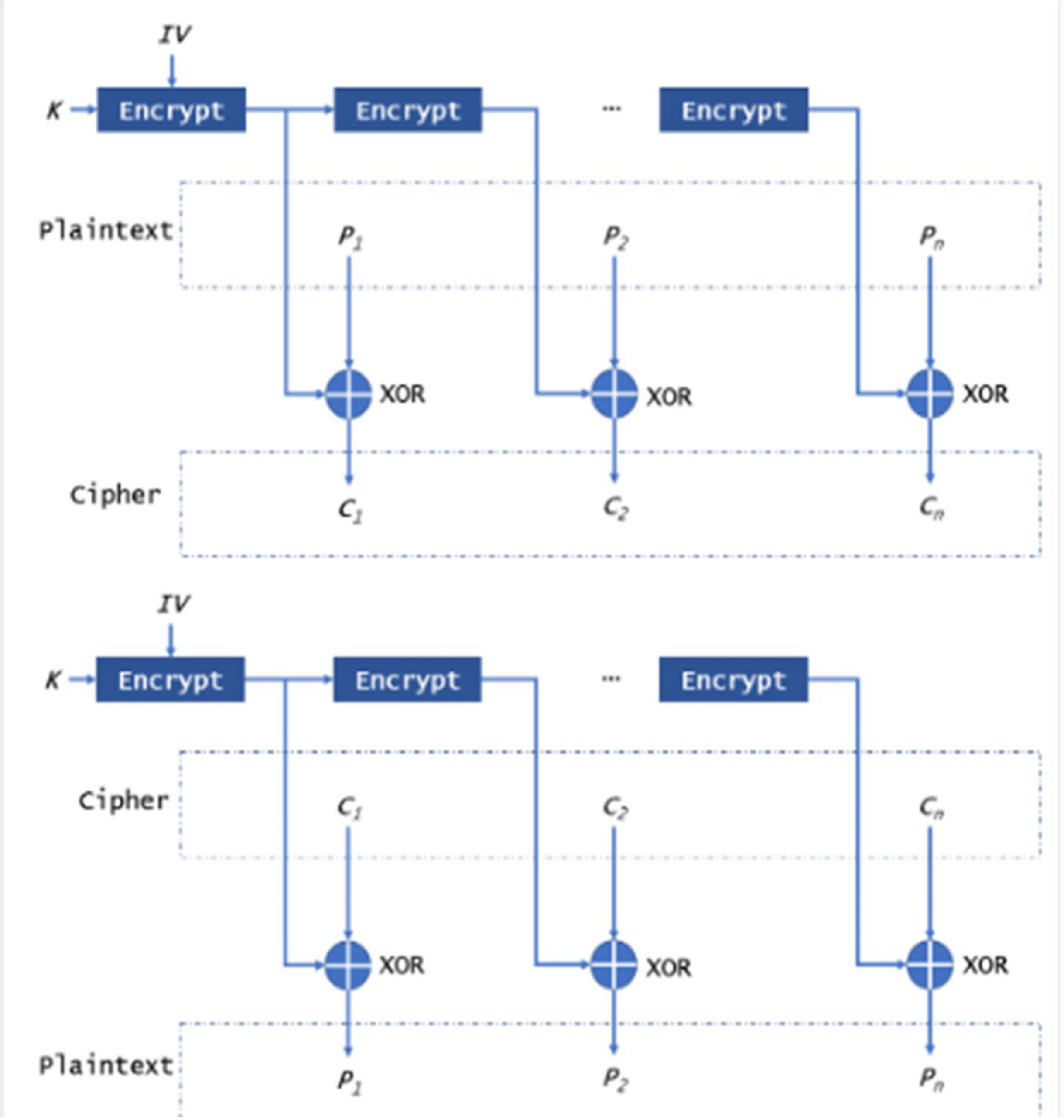


Cipher Feedback (CFB):

- **Description:**
  - CFB transforms a block cipher into a self-synchronizing stream cipher. It operates on plaintext in segments, typically 8 bits at a time (1 byte).
  - The previous ciphertext block is used as the input for the block cipher, and the result is XORed with the plaintext to produce the ciphertext.
- **Pros:**
  - Allows for byte-level encryption, making it suitable for streaming data.
  - Can handle arbitrary lengths of data.
- **Cons:**
  - Error propagation: If a bit is flipped in the ciphertext, it affects the decryption of subsequent blocks.
  - Sequential processing: Encryption and decryption must be done in order.

  Applications:
  - Streaming data encryption: Suitable for encrypting data as it arrives.
  - Real-time communication: Allows encryption and transmission of data in chunks.

The OFB (Output FeedBack) mode

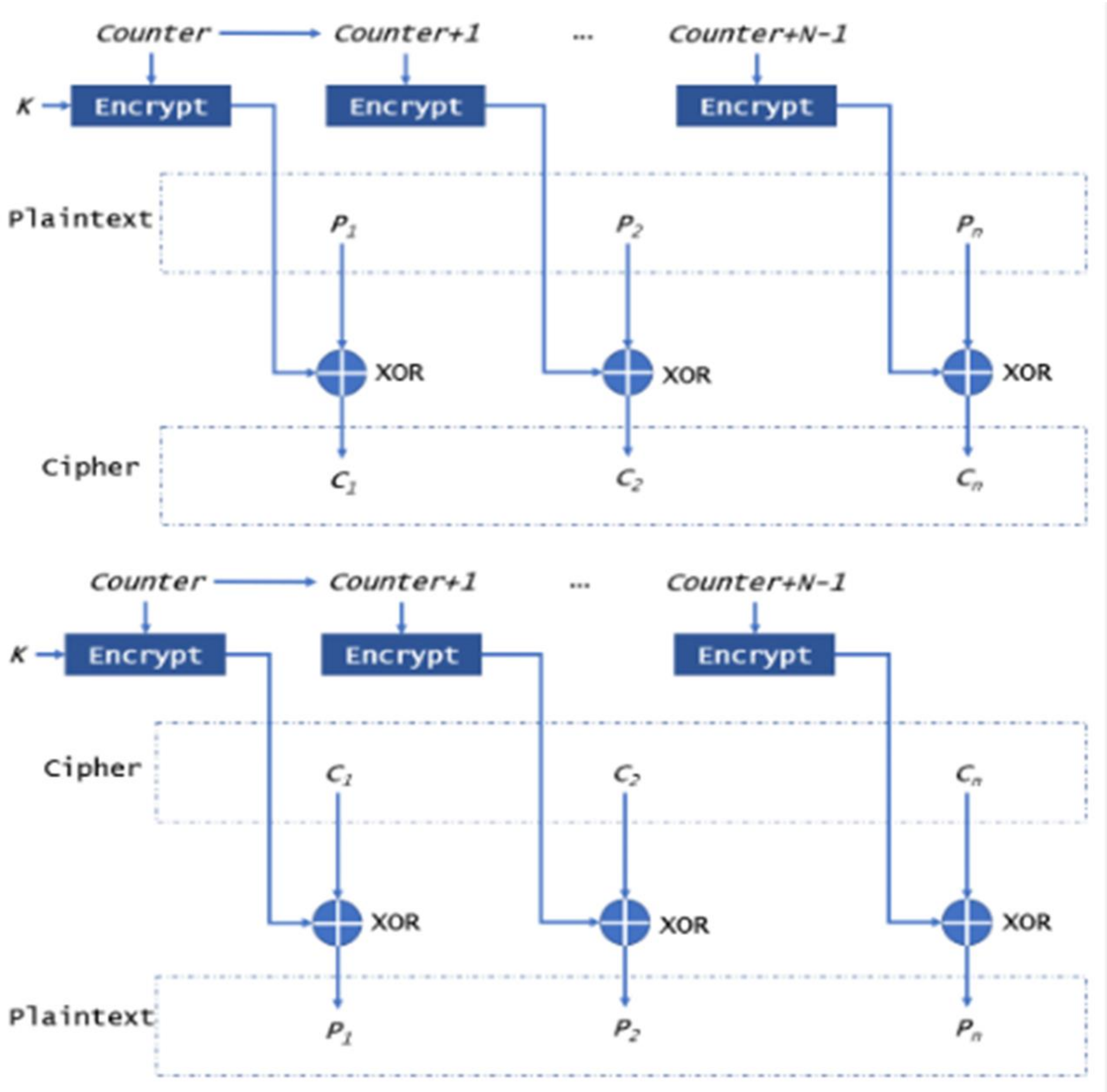


Output Feedback (OFB):

- **Description:**
  - OFB transforms a block cipher into a synchronous stream cipher, generating a keystream independently of the plaintext.
  - The block cipher operates on an initial value (IV), and the output is XORed with the plaintext to produce the ciphertext. The output is then fed back into the block cipher for the next iteration.
- **Pros:**
  - Parallelizable: Multiple blocks can be encrypted simultaneously.
  - Error isolation: If a bit is flipped in the ciphertext, it only affects the corresponding bit in the decrypted plaintext.
- **Cons:**
  - Lack of diffusion: Errors in the ciphertext affect only corresponding bits in the decrypted plaintext.

- **Applications:**
- Stream ciphers: Provides a synchronous stream of pseudorandom bits.
- Secure communication over unreliable channels: Errors in the ciphertext don't affect decryption.

The CTR (Counter) mode



- **Counter (CTR):**
- **Description:**
  - CTR mode turns a block cipher into a stream cipher by using a counter value as input to the block cipher.
  - The counter value is encrypted, and the resulting block is XORed with the plaintext to produce the ciphertext.
- **Pros:**
  - Efficient and easily parallelizable: Each block can be encrypted independently.
  - Random access: Allows direct access to any part of the encrypted data.
- **Cons:**
  - No error propagation detection: Errors in the ciphertext are not detectable during decryption.
  - Nonce/Number used once (Counter) reuse can compromise security: If the same nonce is used with the same key, it can lead to security vulnerabilities.
- **Applications:**

Disk encryption: Allows parallel processing for encrypting data blocks.

Network communication: Efficient for encrypting packets independently.

Comparison of Different Modes

Table 8.1 *Summary of operation modes*

Operation Mode	*Description*	*Type of Result*	*Data Unit Size*
ECB	Each *n*-bit block is encrypted independently with the same cipher key.	Block cipher	n
CBC	Same as ECB, but each block is first exclusive-ored with the previous ciphertext.	Block cipher	n
CFB	Each *r*-bit block is exclusive-ored with an *r*-bit key, which is part of previous cipher text	Stream cipher	$r \leq n$
OFB	Same as CFB, but the shift register is updated by the previous *r*-bit key.	Stream cipher	$r \leq n$
CTR	Same as OFB, but a counter is used instead of a shift register.	Stream cipher	n

Conclusion

Each mode's suitability depends on the specific requirements of the application.

For example, if parallel processing is crucial, CTR mode might be preferred. If error isolation is a priority, OFB may be chosen.

The selection is based on trade-offs between security properties, efficiency, and the desired characteristics for a particular use case.